

Radiant Heating and Cooling Made Simple

Conventional hydronic radiant floor (HRF) heating systems offer significant comfort and efficiency benefits compared to conventional forced air systems. Comfort is improved by delivering heat low, where the people are, and by eliminating forced air noise. Efficiency is improved by reducing indoor temperatures for equal comfort, reducing heat losses associated with thermal stratification and ductwork, and substitution of an energy-efficient pump for the more energy-intensive blower. However, radiant floors currently carry high installation and component costs. In areas where cooling is required, homes with HRF heating usually also need a forced-air cooling system at an additional cost. Combine these issues with home builder concerns about puncturing the radiant tubing during construction, and it's easy to understand why HRF systems are relegated to the high-end, custom-home market.

One solution to this problem is to use the Rapid Radiant Deployment System (RRDS). Developed by Davis Energy Group, the RRDS reduces the costs of HRF systems by standardizing and simplifying the installation process and by using the radiant system to provide cooling and heating. In many cases, this system completely eliminates the cost of a forced-air system.

Features and Benefits

The RRDS combines tools, procedures, and components into a heating-and-cooling system that production builders can use to create comfortable, energy-efficient homes that save occupants money from the day they move in. Radiant heating systems are well known for the superior comfort they provide. In a survey conducted for this project, 97% of occupants of homes using radiant heating systems indicated that they found radiant heating more comfortable than



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forced-air heating. The RRDS also offers simplified installation, with a reduced risk of leaks. Davis Energy Group has identified procedures and tools that simplify and speed up the process of attaching radiant tubing to the steel mesh that reinforces the concrete slab. The RRDS also ensures that the hydronic tubing is held at a consistent depth within the concrete slab, well out of reach of the explosive charge-driven fasteners that are used to attach interior wall plates to the floor. This procedure substantially reduces the risk of puncture and leakage.

There is also the added benefit of positive cash flow for the homeowner. Davis Energy Group modeled the costs and benefits of the RRDS combined

Table 1. Projected Annual Savings

Location	Annual Savings (\$/year)		Mature Market*	
	Current Market	Two-Story	One-Story	Two-Story
Santa Rosa	130	21	266	270
El Toro	69	-47	205	202
Riverside	205	88	341	337
Sacramento	184	76	320	325
Fresno	351	240	487	489
El Centro	1,022 [†]	952 [†]	1,158 [†]	1,324 [†]

Notes: When its incremental costs are included in a 30-year mortgage, the RRDS provides a positive cash flow in most California climates. In extremely hot climates, such as El Centro, additional cooling is required on peak days.

* In a mature market, a negative incremental installed cost is projected for the RRDS, due chiefly to the elimination of forced-air components and the consolidation of heating and water heating systems.

[†] Cooling setpoint may be exceeded.

with an advanced indirect/direct evaporative cooling (IDEC) system in a 1,600 ft² one-story home and a 2,600 ft² two-story home in six California climate zones. Based on current incremental costs for the RRDS, heating energy savings of 15%–30%, and cooling energy savings of 50%–80% will create positive cash flow for the single-story home in

all six climate zones (see Table 1). The analysis also indicates positive cash flow for two-story homes in five of those climate zones. Davis Energy Group expects the RRDS/IDEC system to cost less than conventional HVAC equipment once it has achieved significant production volume and market share.

The RRDS can share an instantaneous gas-fired water heater with an energy factor of 0.82 to provide hot water for high-efficiency space heating and for domestic use. This performance compares favorably to that of a conventional furnace with an annual fuel utilization efficiency (AFUE) of 78%. (These two efficiency values are comparable because the instantaneous water heater has no standby losses.) The system's efficiency also exceeds that of conventional gas water heaters, which have energy factors ranging from 0.5 to 0.7.

The RRDS can be coupled with a ductless, high-efficiency IDEC unit to provide cool air directly to the home during the day. The cooler can also circulate evaporatively cooled water through the radiant system at night to precool the slab and substantially reduce daytime cooling loads. Analysis indicates

that in cooling mode, the system will operate at an effective energy efficiency ratio (EER) exceeding 40, compared with an EER of about 15 for the best available compressor-based equipment.

Applications and Standards

The RRDS/IDEC combination can replace conventional HVAC equipment in new residential construction, particularly in production homes. Modeling by the Davis Energy Group indicates that the combined system can satisfy all heating and cooling loads in all California climate zones except the extremely hot conditions in California's Climate Zone 15. The RRDS/IDEC system is applicable in any region where evaporative cooling is viable and for heating-only applications in all climate zones. The system could also be installed in multifamily and commercial construction where concrete slab floors are often poured on multiple levels.

The RRDS/IDEC system will make it easier for home builders to meet California Title 24 home efficiency requirements. The absence of ductwork eliminates leaks of conditioned air into unconditioned

spaces, as well as losses resulting from ductwork routed through unconditioned spaces.

The RRDS is being applied in four prototype homes in a large subdivision being built in Borrego Springs, California. If the system is successful in these homes, it may be used in all homes built in the subdivision. Davis Energy Group also plans to work with component suppliers to develop a prepackaged hardware center that would minimize on-site plumbing time and facilitate meeting the mature market cost targets used in the economic analyses.



This article was originally developed as an E-Source document through California Energy Commission and Davis Energy Group.

FOR MORE INFORMATION:

To learn more about RRDS, visit www.energy.ca.gov/pier/buildings.